

Assignment 1

Set 1

1) Non repudiation: Sender will send msg to receiver but sender ~~doesn't accept~~ ^{can't deny} that message was sent.

→ How non repudiation achieved using asymmetric cryptography.

i) sender uses private key to create digital signature for message.

This signature acts as proof that sender sent message

ii) Receiver verifies the signature using public key of sender. If signature valid, then message is sent by correct sender.

Ex in banking:

If a person does a transfer to other account, but later tells that he/she did not do the transaction then we can use the signature as proof.

2) Se Impersonate: It allows to use the req things by taking other user's privilege.

Ex: Access database/file with credentials
Web server handle multiple users requests.

web server will do:

i) Fetch user specific data

ii) Details of anything from website.

Security Issues when misused:

Unauthorized access can lead to breaches.

→ If in place of me, someone else uses my bank account it is breach

Set 2:

1) RSA relies on prime numbers.

When 2 prime numbers are multiplied we get a composite number.

But when a composite number has to be broken into 2 prime factors it is difficult (for large numbers)

This is the reason for RSA being strong.

→ Also without knowing private key we can get back the plain text.

Set 3:

1) G-20 summit

↓ Caesar cipher

ROT13 algorithm

ROT 13 = Rotate by 13 positions

apply on
only alphabets

Ans: T-20 f h h z g

2)

In english most frequent letter = E, T, A, O

Least = Z, Q, X, J

Lets assume:

If the most common letter in cipher text = X

Then attacker maps X as E.

No of positions

$$\begin{aligned}\text{shifted} &= X - E \\ &= 24 - 5 \\ &= 19\end{aligned}$$

Do 19 shift for every letter in cipher text to get plain text

Set 4:

1) Attacker knows that original string contains vce, cse, aiml.

Key^{size} used for XOR = 1 byte
 $\leq 8 \text{ bits}$

So for all letter in cipher text if we do xor with key we get plain text.

If in plain text vce, cse, aiml is present then that is the key.

$$\begin{aligned}\text{Key} &= 0x00 \text{ to } 0xFF \\ &= 0 \text{ to } 255\end{aligned}$$

~~def brute force (cipher text):~~

ans: ~~for i in cipher text:~~

Lets assume cipher text is also 8 bits each block.

```
def bruteForce (ciphertext):
```

```
    for i in range(255):
```

```
        plaintext=""
```

```
        for word in ciphertext:
```

```
            plaintext += bytes (word ^ i).decode('utf-8')
```

```
            if "vce" in plaintext or "cse" in plaintext
```

```
                or "aiml" in plaintext:
```

```
                print ("key is", i)
```

```
                break
```

2) CBC uses previously stage cipher text as key which allows to produce diff cipher text for same plain text.

④ Set S

i) ~~Set~~ SetWindowHookEx() → windows API which allows application to capture keyboard stroke without users consent

ii) Providing Admin Privileges

iii) Using Insecure 3rd party software

iv) Lack of encryption

To avoid use:

1) MFA 2) update software

3) Only use principle of least privilege → giving only req permission

4) Antivirus, Firewall

2) In symmetric encryption sharing the same key is problem.
→ bcoz how to securely share the key is difficult when no of users increase

→ In asymmetric:
public key shared openly.
Public key for encryption
and private key for decryption. This helps.

Set 6:

~~1) effects of cache poisoning~~

~~1) DNS cache poisoning:~~

- ~~i) select a ^{DNS} server
ii) send malicious ^{request} ~~response~~ to DNS server.~~

~~These contain fake IP
but looks like trusted.
This will inject~~

Set 5:

1) DNS cache poisoning:

- i) select a DNS server
- ii) When DNS asks for an IP then attacker send malicious IP to DNS which looks like trusted
- iii) Now these are stored in cache
- iv) when others ask for same DNS record then fake IP is only send

Disadvantages / effects:

- 1) Phishing possible
- 2) Malware is sent & spying can be done
- 3) Loss of trust
- 4) DDoS possible
- 5) Data breaches

2) firewall

Set 7:

1) 6 concepts of information assurance:

Confidentiality, Integrity,
Availability, Authenticity,
Authorization, Non Repudiation

Authentication using

Asymmetric Encryption:

write about

Digital Signature

Generation

How to share secret key

using Asymmetric Encryption:

A, B

A → private, public key

B → private, public key.

A, B exchange public keys.

A will encrypt ~~private~~ ^{secret} key
using B's public key
and send to B.

B will decrypt it using private key

Now B will encrypt secret key using A's public key and send to A

A will decrypt it using private key

2] Addressing malicious use of window message

1) MFA 2) Proper validation

Encryption

4) use principle of least privilege

5) Run untrusted application in isolated environment

6) use IDS

Types of malware that attackers can develop using malicious use of window message:

1) Trojan Horse: Helps to run commands on victim computer

2) Dos

3) Keyloggers: Helps to detect which key is pressed.

4) Escalation of privileges: Giving high privileges to run malicious program

Set 8

1) write about DNS cache poisoning

same as set 6 - ①

2) Convert each character to ASCII and then do xor with 0x67

$$A = 65 \quad Z = 90$$

$$a = 97 \quad z = 122$$

$$0 = 48 \quad 9 = 57$$

Given:

Cyber Security

$$0x67 = 103$$

~~$$C = 67 \quad 67 \wedge 103 = 36$$~~

$$C = 67 \quad 67 \wedge 103 = 36 \rightarrow 0x24$$

↓ ↓
decimal hexadecimal

Set 9

book

1) DoS → server not available

DoS done by

- 1) Flooding of request
- 2) Find weakness in system & crash it.

Effects of DoS:

- 1) Service unavailable
- 2) Loss of money
- 3) Resource utilization stopped
- 4) data breach / security breach
- 5) Financial loss

~~Set 10~~